

1) Explain the Concept of Data Warehousing. Discuss its Major Characteristics & imp in decision-making & Business Intelligence.

Def A Data Warehouse is a Centralized System that Stores data collected from different sources for analysis, reporting & decision making.

Major characteristics:-

- Subject-Oriented: Data is organized by Subject like Customers, Sales, & products.
- Integrated: Data from different sources is combined in a Common format.
- Time Variant: Historical data is stored for Comparison & analysis.
- Non-Volatile: Stored data is mainly used for reading & analysis.
- Centralized: Data is maintained in one Central repository.

3) Importance

Data Warehousing helps organization

- Identify trends & patterns.
- Generate accurate reports.
- Analyze historical data.
- Support better business decision.
- Provide data for Business Intelligence (BI) tool & dashboards.

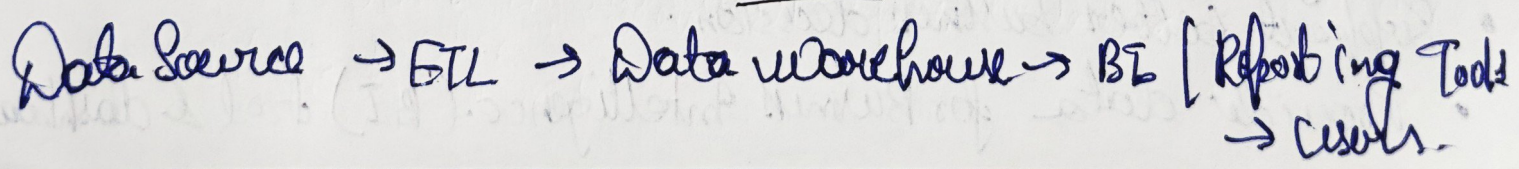
Q2) Differentiate b/w an operational Database & a Data Warehouse based on purpose, data characteristics, user, processing & update frequency?

Ans Basic	operational Database	Data Warehouse
Purpose	Handle daily business transaction	used for analysis & decision making
Data	Current & detailed data	Historical & Summarized data
Users	Employee, Customer & application	Manager, analysts & decision making
Processing	Mainly OLTP (transaction processing)	Mainly OLAP (analytical processing)
Updates	updates frequently & continuously	updates periodically usually through ETL

Q3 Explain the architecture of a Data Warehouse with its major components. Describe the role of the ETL process in the architecture

Ans A data warehouse architecture show how data move from different source to the warehouse & finally to users for analysis.

Architecture



2) Major Components -

- Data Sources: Databases, application, files & external system
- ETL Process: Extract, clean, transform & load data
- Data Warehouse - Central storage for integrated & historical data
- Data Marts: Smaller section of the warehouse for specific department.
- BI Tools: used for reports, dashboard & data analysis.

3. Role of ETL

ETL stands for Extract, Transform, Load.

- Extract: Collects data from different source
- Transform: cleans & converts data into a common format
- Load: store the transform data in the data warehouse

4) A retail organization collects sale data from multiple branches, customer databases & online transactions. Design a suitable Data warehouse architecture for the organization & explain the flow of data from source system to reporting.

1) Proposed Architecture

Branch Sales → Customer Database & Online Transactions
 → ETL → Data Warehouse → BI/Reporting → Management

2) Dataflow

Global Aggregates

- 1) Source System: Sales data from branch, Customer databases & online transactions is collected
- 2) ETL Process: Data is Extracted, Transformed & Cleaned into a Common format
- 3) Data Warehouse: Clean & integrated data is stored centrally with historical record.
- 4) Data Marts: Separate data centres provided for area like sales, Customer & inventory
- 5) Reporting / BI: BI tools use warehouse data to create reports, dashboards & charts
- 6) Management: Managers analyze the reports to make better business decision
- 7) Consider a Sales Data Warehouse containing Product, Customer, Time & Sales dimension. Explain how OLAP operation such as Roll-up, Drill-down, Slice, Dice & Pivot can be applied to analyze sales. Give a suitable example for each operation.

Ans A Sales Data Warehouse contains four ^{Global Aggregates} main dimensions Product, Customer, Time & Sales. OLAP (Online Analytical Processing) operation helps analyze these data from different views.

1) Roll-up → Roll-up summarizes data by moving from a lower level to a higher level.

Example: Daily Sales → Monthly Sales → Yearly Sales

2) Drill-down.

Drill-down is the opposite of roll-up. It provides more detailed information.

Example: yearly sale → Monthly sale → Daily sales

3) Slice - Slice selects a single value one dimension → to create a smaller view of the data

Example: Selecting 2026 from the time dimension to create a smaller view of the data -
only 2026 sales

4) Dice Dice selects multiple values from two or more dimensions.

Example: - Analyzing sales of laptop & Mobile for customers in Delhi & Noida during 2026

5) Pivot : Pivot changes the orientation of the data to provide a different view for analysis

Example : Sales can be viewed as Product by Month instead of Month by Product.